



NATIONAL SCHOOL OF BUSINESS MANAGEMENT

BSc in Management Information Systems (Special) – 16.2/17.1

4th Year 2nd Semester Examination

1 November 2020

IS404.3 - Business Analytics

Instructions to Candidates

- 1) Answer only five (05) questions**
- 2) Time allocated for the examination is three and half (3.5) hours.
- 3) Calculators are allowed.
- 4) Total number of pages - Eight (08)
- 5) Use the provided answer booklet to write your answers and upload to the LMS as a PDF file within the stipulated time. This PDF file should be saved with your index no.
- 6) You are required to submit the answer file within the allowed timeframe, and no additional time is provided for submissions.
- 7) All the submissions will be checked for plagiarism. Plagiarism, collusion, and copying are serious offences in the university and serious penalties that would be imposed.
- 8) You're not allowed to publish or disseminate any part of the paper online or offline.

Question 1**[20 marks]**

- a) Consider the problem of producing hand sanitizer, which kills germs and destroys Corona Virus. This sanitizer contains two active ingredients. One liter (1 L) of sanitizer liquid must contain a minimum quantity of each of four chemicals as below:

Chemical	A	B	C	D
Minimum requirement in milliliters	90	50	20	2

The ingredients have the following nutrient values and cost:

	A	B	C	D	Cost/L
Ingredient 1 (ml/L)	100	80	40	10	40
Ingredient 2 (ml/L)	200	150	20	-	60

Formulate this problem as an LPP and use a graphical method to find, what should be the amount of active ingredients in one liter of hand sanitizer in order to reduce the cost?

[10 marks]

- b) Use simplex method to solve the following linear programming problem.

$$\text{Minimize } Z = 5X_1 + 2X_2 + 4X_3$$

Subject to

$$3X_1 + X_2 + 2X_3 \geq 4$$

$$6X_1 + 3X_2 + 5X_3 \geq 10$$

$$X_1, X_2, X_3 \geq 0$$

[10 marks]

Question 2**[20 marks]**

- a) Explain Bayes' Theorem in your words using an example (You may use diagrams and equations).
[5 marks]
- b) A certain illness X has only one of the two symptoms A and B. it is known that (in the usual notations) $P(X|A) = 0.2$ and $P(X|B) = 0.8$.
In a certain population 40% have symptom A and the remaining 60% have the symptom B.
Calculate the probability that a randomly selected person has illness X.
Also show that the probability of symptom B being shown, given that a patient is suffering from the illness X is $6/7$.
[5 marks]
- c) Suppose you need to construct a Naïve Bayes Classifier to separate a given day as a Sunny day or Rainy day. If it is possible to measure features, Temperature in $^{\circ}\text{C}$ and Humidity in % of every day, explain how you would achieve this task. Also state any assumption you made.
[5 marks]
- d) Calculate the probabilities and Classify a new day with feature Temperature = 27.5°C and Humidity = 68%

Sunny Day					
Temperature	Frequency	Probability	Humidity	Frequency	Probability
23 – 24	0		10-20	10	
24 – 25	0		20-30	15	
25 – 26	0		30-40	35	
26 – 27	15		40-50	40	
27 – 28	20		50-60	30	
28 – 29	30		60-70	20	
30 – 31	40		70-80	0	
31 – 32	25		80-90	0	
32– 33	20		90-100	0	

Rainy Day					
Temperature	Frequency	Probability	Humidity	Frequency	Probability
23 – 24	20		10-20	0	
24 – 25	25		20-30	0	
25 – 26	40		30-40	0	
26 – 27	30		40-50	20	
27 – 28	20		50-60	25	
28 – 29	15		60-70	35	
30 – 31	0		70-80	40	
31 – 32	0		80-90	20	
32– 33	0		90-100	10	

[5 marks]

Question 3**[20 marks]**

Business Analyst of a clothing company was given a task to evaluate and decide on a marketing plan for their new product. Cost of Plan A will be \$500,000 while a more conservative approach, Plan B, will cost only \$100,000. Table shows the projected gross profits (profits before advertising) on the new product for each advertising plan under two possible states of nature which are complete acceptance of the new product and limited acceptance. Both the scenarios are equally likely.

Advertising Plan	State of Nature	
	Limited Acceptance	Complete Acceptance
Plan A	\$400,000	\$1,000,000
Plan B	\$500,000	\$500,000

BA estimates that there are equal chances of complete and limited acceptance of the new product

- a) Set up a Net profit payoff matrix.(Net profit = Gross profit – Cost) [1 mark]
- b) Find the optimal plan using decision making criteria under uncertainty. [5 mark]
- c) Use BA's subjective probability estimates to choose an advertising plan based on Expected Value Criterion. [2 marks]
- d) What is the value of perfect information in this situation? [2 marks]
- e) Before launching the product, a survey costing \$50,000 can be run to test -market the product. In past uses, this survey has been shown to predict complete acceptance in 60% of the cases where there was acceptance and predicted limited acceptance 70% of the time when there was limited acceptance. Use this information to determine whether this survey should be used to help decide on an advertising plan. [5 marks]
- f) Use a decision tree to show your analysis. [5 marks]

Question 4**[20 marks]**

- a) For an election two political parties A and B are planning of nominating a candidate and nominations are to be announced simultaneously. Senuki, Sarath and Sudesh represent party A and Singh, Savi and Banu represent party B.

Payoff table for Party A is given below.

		Party B		
		Singh	Savi	Banu
Party A	Senuki	3/4	4/5	1/4
	Sarath	3/10	3/5	1/5
	Sudesh	4/5	1/5	1/4

Develop the payoff table for Party B and show that this is a zero-sum game.

[5 marks]

- b) Solve the following game. (The payoffs with respect to player A is given in the table)

		Player B			
		Y1	Y2	Y3	Y4
Player A	X1	19	6	7	5
	X2	7	3	14	6
	X3	12	8	18	7
	X4	8	7	13	-1

[5 marks]

- c) Solve the following game using graphical approach

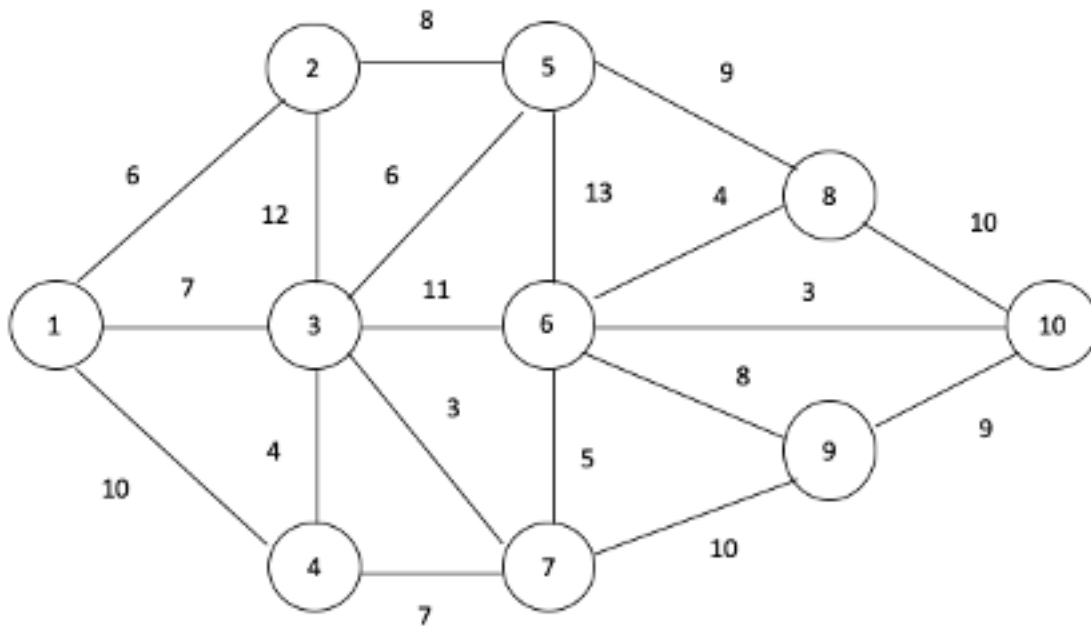
		Player B	
		B1	B2
Player A	A1	-2	0
	A2	3	-1
	A3	-3	2
	A4	5	-4

[10 marks]

Question 5

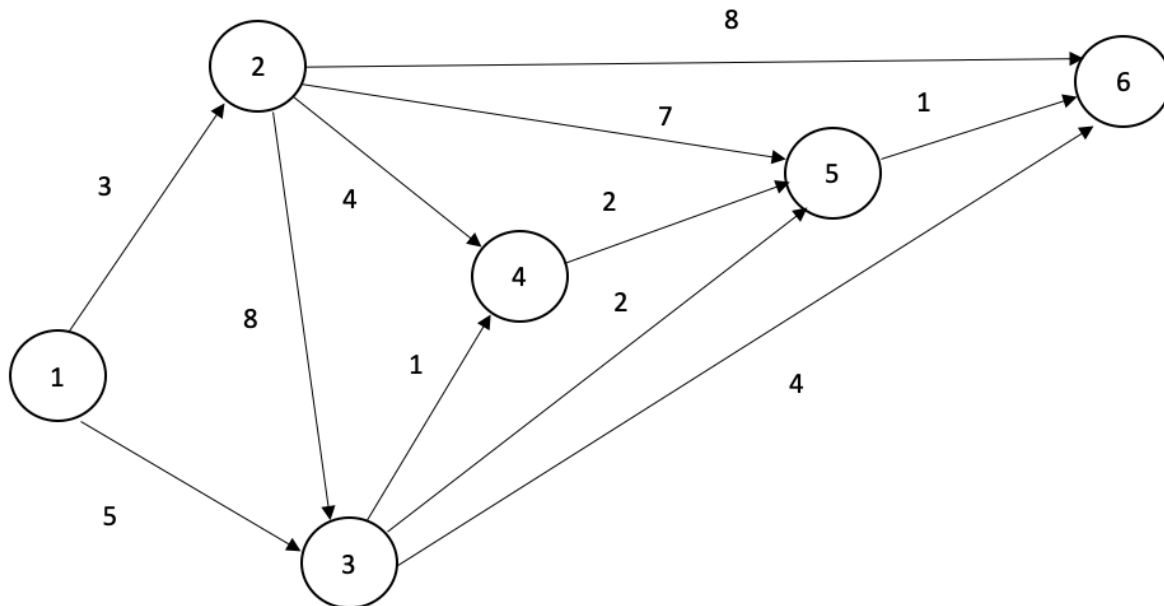
[20 marks]

- a) Consider the road network given below where the distance between different pairs of adjacent cities is summarized. Find the minimum spanning tree using Kruskal's algorithm.



[10 marks]

- b) Find the shortest path between node 1 to 6 in the following network using Dijkstra's Algorithm.



[10 marks]

Question 6**[20 marks]**

Winning average speeds (in miles per hour) of a contestant in a certain game which happens in every five years are shown in the following table.

Year	1965	1970	1975	1980	1985	1990	1995	2000	2005
Speed	109	114	116	117	114	127	131	138	141

- a) Plot these data and comment on their suitability for simple linear regression analysis.
- b) Calculate and interpret the Pearson's correlation coefficient.
- c) What goodness-of fit criterion is used to determine the regression coefficients? Briefly explain.
- d) Fit a simple linear regression model in usual notations
- e) Draw the line of best fit on the same plot.
- f) Compute the Coefficient of Determination and assess the model.
- g) Why we need to assess the model?
- h) It is later decided that driving in 1985 were affected by flooding of the track. Therefore, the values in the year 1985 omitted and the regression was recalculated.

The regression equation is $y = -1464 + 0.800 x$

$$R^2 = 97.9\%$$

Is the removal of observation at 1985 recommended? Justify your answer.

- i) Use the best model which you recommend and find the expected winning speed in 2010.
- j) According to the same model by which time winning speed of 160 mph might be expected?

[2 marks each]

*End of Paper *